

CIRCLE

CHAPTER

2

"Every body of discovery is mathematical in form because there is no other guidance we can have."

– Darwin

2.1 INTRODUCTION

We come across objects of different shapes in our daily life. For example, the shape of a stick is a straight line, the form of a class room's floor is a rectangle. In the preceding chapter, we have studied various forms of the equations of a straight line. In this chapter, we shall study about the curve known as circle and in the forthcoming chapters some other curves known as parabola, ellipse and hyperbola are introduced.

Examples of circle are denominations of ₹1, ₹5 coins, buttons of shirts, round dials of clocks. Also the shape of a wheel of a motor cycle or a bullock-cart and that of bangle is a circle. This curve has a wide range of applications in Physics, Chemistry and Engineering. For example, we find the temperature at a point inside the circle of radius 'r' subject to certain conditions.

2.2 LOCUS

The path traced out by a point when it moves in space, under given conditions or in accordance with a definite law, is known as a locus of that point (Loci is the plural of locus).

Read the following examples :

1. If an animal is tied with a rope to a post in the ground, what path will it describe, if it moves so as always to keep the rope taut?

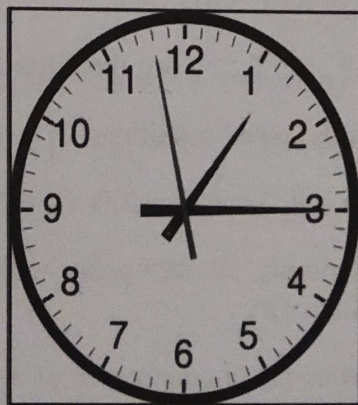
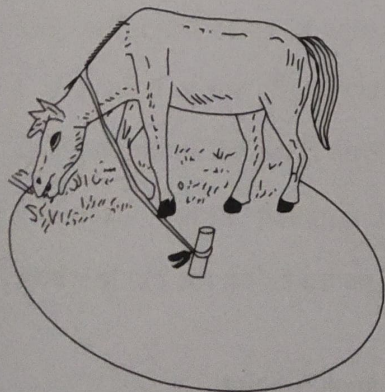


Fig. 2.1

2. If you look at the tip of the seconds hand of a watch, you see that it completes one revolution in one minute and that in completing this revolution it traces out the circumference of a circle. Will the tip always describes the same path? If, so why?

It is clearly seen that

(a) In the first case, the animal moves along a circle if

(i) the post is fixed,

(ii) the animal moves so that the rope remains taut. That is, its distance from the post does not change and it remains constant.

Had these two conditions not been imposed on the motion of the animal, it would not have moved along a fixed path (the circle).

(b) The tip of the seconds hand describes a circle because

(i) its one end is fixed,

(ii) when it moves round the centre, the distance between the two ends remains constant. If the tip of the seconds hand of the watch were not bound to fulfil these conditions it would not have described a fixed path (circle).

Definition :

The path of a moving point under a given condition (or conditions) is called the locus of the point. It may be a straight line or a curve.

It is seen from the above definition of locus that

(i) Every point lying on that line or curve (locus) will satisfy the given conditions.

(ii) Every point satisfying the given conditions will lie on that line or curve.

Illustrations :

Condition : Let O be a fixed point in a plane and a point P moves in the plane such that its distance from the fixed point remains constant ($= a$).

Locus : $P_1, P_2, P_3, \dots$ are the different positions of P.

By the given condition,

$$PO = P_1O = P_2O = P_3O = \dots$$

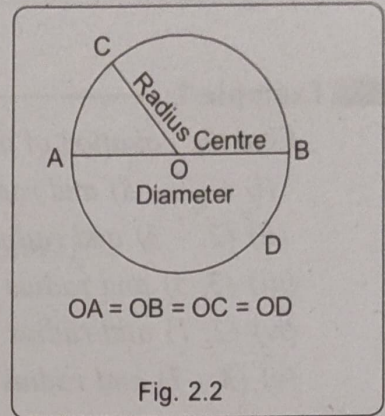
Obviously, all such points will be on the circle as a path i.e., on the circle whose centre is O and radius OP.

Hence the circumference of a circle is the locus of such a point.

2.3 CIRCLE

Definition : A circle is the locus of a point which moves in a plane such that its distance from a fixed point is always constant. The fixed point is called the **centre** and the constant distance is called the **radius** of the circle. The word radius comes from the Latin word for rod or spoke of a wheel, and the radii, or spokes, radiate out from the centre.

- Note :**
1. If the radius of a circle is unity then it is called a **unit circle**.
 2. If the radius of a circle is zero then it is called a **point circle**.



2.4 EQUATION OF THE CIRCLE WHOSE CENTRE IS (h, k) AND RADIUS r IS $(x - h)^2 + (y - k)^2 = r^2$

Proof : Let OX and OY be the two rectangular axes. Let C be the centre of the circle and P(x, y) is any point on the circle.

Then $CP = r$

Squaring both sides, $CP^2 = r^2$

$$\text{i.e., } (x - h)^2 + (y - k)^2 = r^2 \quad \dots (1)$$

This is the required equation of the circle.

Simplifying the equation (1), it reduces to

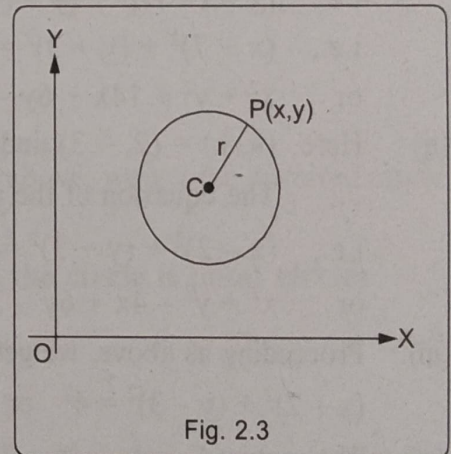
$$x^2 + y^2 - 2hx - 2ky + h^2 + k^2 - r^2 = 0$$

which is of the form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

It is noticed from this equation that

1. It is a second degree equation in x and y.
2. The coefficients of x^2 and y^2 are equal
3. The coefficient of xy is zero.



- Note :**
1. Equation of the circle whose centre is the origin and radius r is $x^2 + y^2 = r^2$.
 2. Equation of the point circle whose centre (h, k) is $(x - h)^2 + (y - k)^2 = 0$.
 3. Equation of the point circle whose centre (0, 0) is $x^2 + y^2 = 0$.

Solved Examples

Example 1:

Find the equation of the circle, whose centre is the point

(i) $(-7, -3)$ and radius is 10.

(ii) $(2, -3)$ and radius 4.

(iii) $(2, 3)$ and radius 4.

(iv) $(2, 1)$ and radius 3.

(v) $(3, -7)$ and radius 5.

[April 2015 (D.C.E)]

[April 2014 (D.E.C.E), April 2017 (D.M.E), 2018 (D.E.E.E)]

[April 2016 (D.E.C.E)]

Solution :

(i) Given centre is $(-7, -3)$

i.e., $(h, k) = (-7, -3) \Rightarrow h = -7, k = -3$

and radius is $r = 10$.

$\therefore$ The equation of the circle is $(x - h)^2 + (y - k)^2 = r^2$

i.e., $[x - (-7)]^2 + [y - (-3)]^2 = 10^2$

i.e., $(x + 7)^2 + (y + 3)^2 = 100$ i.e., $x^2 + 14x + 49 + y^2 + 6y + 9 = 100$

or $x^2 + y^2 + 14x + 6y - 42 = 0$. This is the equation of the required circle.

(ii) Here $(h, k) = (2, -3)$ and $r = 4$

$\therefore$ The equation of the required circle is $(x - h)^2 + (y - k)^2 = r^2$

i.e., $(x - 2)^2 + (y + 3)^2 = 4^2$ (or) $x^2 - 4x + 4 + y^2 + 6y + 9 = 16$

or $x^2 + y^2 - 4x + 6y - 3 = 0$.

(iii) Proceeding as above, we get the equation of the required circle as

$(x - 2)^2 + (y - 3)^2 = 4^2$ or $x^2 + y^2 - 4x - 6y - 3 = 0$.

(iv) The required circle is $(x - 2)^2 + (y - 1)^2 = 3^2$ or $x^2 + y^2 - 4x - 2y - 4 = 0$.

(v) Proceeding as above, we get the required circle as $x^2 + y^2 - 6x + 14y + 33 = 0$.

Example 2:

Find the equation of the circle whose radius is $\sqrt{a^2 - b^2}$ and whose center is $(-a, -b)$.

[ECET 2014]

Solution :

Given radius, $r = \sqrt{a^2 - b^2} \Rightarrow r^2 = a^2 - b^2$

and center, $C = (-a, -b)$ i.e., $(h, k) = (-a, -b) \Rightarrow h = -a, k = -b$

Hence the equation of the required circle is $(x - h)^2 + (y - k)^2 = r^2$

i.e., $[x - (-a)]^2 + [y - (-b)]^2 = a^2 - b^2$

i.e., $(x + a)^2 + (y + b)^2 = a^2 - b^2$

i.e., $x^2 + 2ax + a^2 + y^2 + 2by + b^2 = a^2 - b^2$

or $x^2 + y^2 + 2ax + 2by + 2b^2 = 0$.

Example 3:

Find the equation of the point circle with centre

- (i) $(-7, 4)$ [April 2018 (D.E.C.E.)]
 (ii) $(7, -9)$
 (iii) $(-5, 2)$
 (iv) $(5, -2)$ [April 2018 (Common), Jan 2023 (MID-I), April 2023]
 (v) $(1, 2)$ [April 2017 (A/AA/AEI...)]
 (vi) $(2, 3)$

Solution :

- (i) We know that the equation of the circle having the centre at (h, k) and radius r is
 $(x - h)^2 + (y - k)^2 = r^2$
 Here $h = -7$, $k = 4$ and radius, $r = 0$. (Since the circle is point circle)
 $\therefore$ The equation of the required circle is
 $(x + 7)^2 + (y - 4)^2 = 0$ or $x^2 + y^2 + 14x - 8y + 65 = 0$
- (ii) Proceeding as above, we obtain the required circles as
 $x^2 + y^2 - 14x + 18y + 130 = 0$.
- (iii) This is left as an exercise to the reader. Proceeding as the above, we get the required circle as $x^2 + y^2 + 10x - 4y + 29 = 0$.
- (iv) Given centre $= (h, k) = (5, -2)$ and radius, $r = 0$. (Since the circle is point circle)
 $\therefore$ Equation of the required circle is
 $(x - h)^2 + (y - k)^2 = r^2$
 i.e., $(x - 5)^2 + (y + 2)^2 = 0$
 i.e., $x^2 - 10x + 25 + y^2 + 4y + 4 = 0$
 or $x^2 + y^2 - 10x + 4y + 29 = 0$.
- (v), (vi) This is left as an exercise to the reader.

Example 4:

Find the equation of the circle whose centre is $(3, -4)$ and which passes through the origin.

Solution :

Let $C(3, -4)$ be the centre, and r be the radius of the circle

Let $P(0, 0)$ be the origin.

Then $CP = r$

i.e. $CP^2 = r^2$ i.e., $(3 - 0)^2 + (-4 - 0)^2 = r^2$

i.e., $9 + 16 = r^2$ or $r = 5$

$\therefore$ The equation of the circle with centre $(3, -4)$ and radius 5 is

$$(x - 3)^2 + (y + 4)^2 = 5^2$$

$$\text{i.e., } x^2 - 6x + 9 + y^2 + 8y + 16 = 25$$

$$\text{or } x^2 + y^2 - 6x + 8y = 0$$

An Easiest Method.

Let the required circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \dots (1)$$

Since it is passing through the origin,

$$\therefore c = 0.$$

Given centre of the required circle is $(3, -4)$

$$\text{i.e. } -g = 3 \text{ and } -f = -4 \text{ (Refer article 2.3)}$$

$$\text{or } g = -3 \text{ and } f = 4$$

Substituting the values of g, f and c in (1), we get the desired circle as $x^2 + y^2 - 6x + 8y = 0$

Example 5 :

Find the equation of the circle which has its centre at the point $(2, 1)$ and touches the straight line $3x + 4y - 5 = 0$.

Solution :

The radius of the circle is the perpendicular distance from the centre to the straight line $3x + 4y - 5 = 0$.

$$\therefore r = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{|3(2) + 4(1) - 5|}{\sqrt{3^2 + 4^2}} = 1$$

Hence the equation of the circle is

$$(x - 2)^2 + (y - 1)^2 = (1)^2$$

$$\text{i.e., } x^2 + y^2 - 4x - 2y + 4 = 0$$

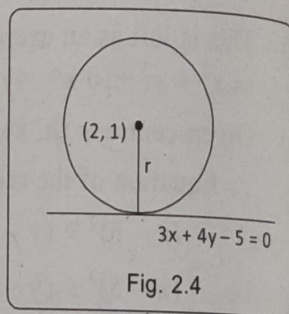


Fig. 2.4

2.5 THE EQUATION $x^2 + y^2 + 2gx + 2fy + c = 0$ ALWAYS REPRESENTS A CIRCLE

Proof : The equation can be written as $x^2 + y^2 + 2gx + 2fy = -c$

Adding $g^2 + f^2$ on both sides, we get

$$(x^2 + 2gx + g^2) + (y^2 + 2fy + f^2) = g^2 + f^2 - c$$

$$\text{i.e., } (x + g)^2 + (y + f)^2 = [g^2 + f^2 - c]$$

$$\text{i.e., } [x - (-g)]^2 + [y - (-f)]^2 = [\sqrt{g^2 + f^2 - c}]^2$$

This is of the form $(x - h)^2 + (y - k)^2 = r^2$

Hence the equation $x^2 + y^2 + 2gx + 2fy + c = 0$ always represents a circle whose centre is $(-g, -f)$ and radius is $\sqrt{g^2 + f^2 - c}$.

Note 1: (i) If $g^2 + f^2 - c > 0$, the radius of the circle is real. So the equation represents a real circle whose centre is $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$.

(ii) If $g^2 + f^2 - c = 0$, the radius vanishes and the equation of the circle becomes $(x + g)^2 + (y + f)^2 = 0$.

This is called a **point circle** or **zero circle** and the only real solution of the equation is $x = -g, y = -f$. In this case the circle reduces to the point $(-g, -f)$.

(iii) If $g^2 + f^2 - c < 0$, the radius of the circle is imaginary. So the equation represents an imaginary circle.

Note 2 : The general equation of the second degree in x and y represents a circle if the coefficients of x^2 and y^2 are equal and if the coefficient of xy be zero.

i.e., The equation $ax^2 + ay^2 + 2gx + 2fy + c = 0$ represents a circle.

Note 3 : The centre of the circle $ax^2 + ay^2 + 2gx + 2fy + c = 0$ is $\left(\frac{-g}{a}, \frac{-f}{a}\right)$ and radius is

$$\frac{1}{a} \cdot \sqrt{g^2 + f^2 - ac}.$$

2.6 EQUATION OF THE CIRCLE HAVING (x_1, y_1) AND (x_2, y_2) AS ENDS OF A DIAMETER IS $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

Proof : Let A be the point (x_1, y_1) and B be the point (x_2, y_2) and let $P(x, y)$ be any point on the circle. Join AP and BP.

Since the angle in a semi-circle is a right angle,

$$\therefore \angle APB = 90^\circ$$

$$\text{Slope of AP} = m_1 = \frac{y - y_1}{x - x_1}.$$

$$\text{Slope of BP} = m_2 = \frac{y - y_2}{x - x_2}$$

$$\text{Since } AP \perp BP, m_1 \cdot m_2 = -1$$

$$\text{i.e., } \frac{y - y_1}{x - x_1} \cdot \frac{y - y_2}{x - x_2} = -1$$

$$\text{i.e., } (x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

This therefore is the required equation.

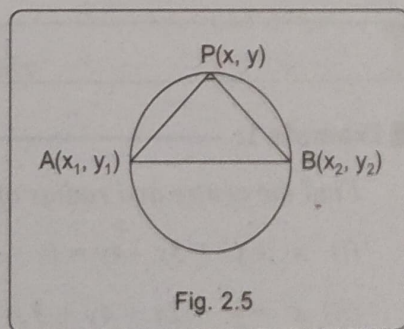


Fig. 2.5

Note : If (x_1, y_1) is one end of a diameter of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, then the other end is $(-2g - x_1, -2f - y_1)$.

2.7 DEFINITIONS

1. **Concentric circles.** Circles having the same centre are called concentric circles.
2. **Equal circles.** Circles having equal radius are called equal circles.

2.8 EQUATION OF CIRCLE GIVEN CENTRE AND A POINT ON THE CIRCUMFERENCE

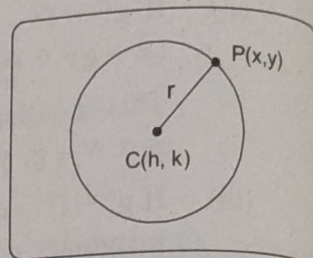
Definition : The length of the complete circle is called its circumference. In fact, it is the distance around a circle.

Let $C(h, k)$ be the centre of the given circle and $P(x, y)$ be the given point on the circumference. Also let r be the radius of the circle. Then we have

$$CP = r \Rightarrow CP^2 = r^2$$

$$\Rightarrow (x - h)^2 + (y - k)^2 = r^2$$

which is the equation of the required circle.



2.9 EQUATION OF CIRCLE THROUGH THREE NON-COLLINEAR POINTS

We know that two points are sufficient to determine a line. That is, there is one and only one line passing through two points. Now, a natural question arises : how many points are sufficient to determine a circle ? The answer to this question is that three non-collinear points are sufficient to determine a circle.

Result : A unique circle passes through three non-collinear points.

To find the equation of the circle through three given non-collinear points, take the general equation of the circle as $x^2 + y^2 + 2gx + 2fy + c = 0$.

Substitute given three points in the equation and get three corresponding equations in g , f and c . Solve them to get the values of g , f and c and then substitute these values in the general equation to obtain the required circle.

Solved Examples

Example 1:

Find the centre and radius of the circle

(i) $x^2 + y^2 + 3x - 4y = 0$.

[April 2013, 2014 (D.E.E.E), April 2018 (D.E.C.E)]

(ii) $x^2 + y^2 + 2x + 4y + 5 = 0$

[April 2016 (D.M.E)]

(iii) $x^2 + y^2 + 4x + 6y + 5 = 0$

[April 2016 (A/AA/AEI...), April 2017 (D.M.E)]

Solution :

- (i) Comparing the given circle with the standard form

$$x^2 + y^2 + 2gx + 2fy + c = 0, \text{ we have}$$

$$2g = 3, 2f = -4 \text{ and } c = 0 \Rightarrow g = \frac{3}{2} \text{ and } f = -2$$

$$\therefore \text{Centre} = (-g, -f) = \left(-\frac{3}{2}, 2\right)$$

$$\begin{aligned} \text{and Radius} &= \sqrt{g^2 + f^2 - c} = \sqrt{\left(\frac{3}{2}\right)^2 + (-2)^2 - 0} \\ &= \sqrt{\frac{9}{4} + 4} = \sqrt{\frac{9+16}{4}} = \sqrt{\frac{25}{4}} = \frac{5}{2} \end{aligned}$$

(ii), (iii) These are left as an exercise to the reader.

Example 2:

Find the centre and radius of the circle

(i) $3x^2 + 3y^2 - 5x - 6y + 4 = 0.$

[April 2018 (Common)]

(ii) $3x^2 + 3y^2 - 11x - 7y + 1 = 0.$

[April 2010 (D.E.C.E), Oct. 2011 (D.M.E),
April 2017 (D.C.E)]

(iii) $27x^2 + 27y^2 + 6x - 8y - \frac{2}{3} = 0.$

Solution :

- (i) Given circle is $3x^2 + 3y^2 - 5x - 6y + 4 = 0$. This equation is not in the standard form of the circle i.e., $x^2 + y^2 + 2gx + 2fy + c = 0$. To reduce it into the standard form, we divide the given equation by 3.

Dividing by 3, we get

$$x^2 + y^2 - \frac{5}{3}x - 2y + \frac{4}{3} = 0$$

Comparing it with $x^2 + y^2 + 2gx + 2fy + c = 0$, we have

$$2g = -\frac{5}{3}, 2f = -2, c = \frac{4}{3}$$

$$\text{i.e., } g = -\frac{5}{6}, f = -1 \text{ and } c = \frac{4}{3}$$

$$\therefore \text{The centre is } (-g, -f) \text{ i.e., } \left(\frac{5}{6}, 1\right)$$

$$\text{Radius} = \sqrt{g^2 + f^2 - c} = \sqrt{\frac{25}{36} + 1 - \frac{4}{3}} = \sqrt{\frac{25+36-48}{36}} = \frac{\sqrt{13}}{6}$$

Aliter : Comparing the given equation with

$$ax^2 + ay^2 + 2gx + 2fy + c = 0, \text{ we have } a = 3, 2g = -5, 2f = -6, c = 4$$

$$\Rightarrow a = 3, g = \frac{-5}{2}, f = -3, c = 4$$

$$\therefore \text{Centre} = \left(\frac{-g}{a}, \frac{-f}{a} \right) = \left(\frac{5}{6}, \frac{3}{5} \right) = \left(\frac{5}{6}, 1 \right)$$

$$\text{Radius} = \frac{1}{a} \cdot \sqrt{g^2 + f^2 - ac} = \frac{1}{3} \cdot \sqrt{\frac{25}{4} + 9 - 12} = \frac{1}{3} \cdot \sqrt{\frac{25}{4} - 3} = \frac{1}{3} \sqrt{\frac{13}{4}} = \frac{\sqrt{13}}{6}$$

(ii) Dividing the equation of the circle by 3, we get

$$x^2 + y^2 - \frac{11}{3}x - \frac{7}{3}y + \frac{1}{3} = 0$$

Comparing it with $x^2 + y^2 + 2gx + 2fy + c = 0$, we have

$$2g = \frac{-11}{3}, 2f = \frac{-7}{3} \text{ and } c = \frac{1}{3} \Rightarrow g = \frac{-11}{6}, f = \frac{-7}{6} \text{ and } c = \frac{1}{3}$$

$$\therefore \text{Centre of the circle} = (-g, -f) = \left(\frac{11}{6}, \frac{7}{6} \right)$$

$$\text{Radius of the circle} = \sqrt{g^2 + f^2 - c} = \sqrt{\left(\frac{11}{6} \right)^2 + \left(\frac{7}{6} \right)^2 - \frac{1}{3}} = \sqrt{\frac{121}{36} + \frac{49}{36} - \frac{1}{3}}$$

$$= \sqrt{\frac{121 + 49 - 12}{36}} = \frac{1}{6} \cdot \sqrt{158}$$

(iii) Comparing the given equation with $ax^2 + ay^2 + 2gx + 2fy + c = 0$, we have $a = 27, 2g = 6$

$$2f = -8, c = \frac{-2}{3} \Rightarrow a = 27, g = 3, f = -4, c = \frac{-2}{3}$$

$$\therefore \text{Centre} = \left(\frac{-g}{a}, \frac{-f}{a} \right) = \left(\frac{-3}{27}, \frac{4}{27} \right)$$

$$\text{Radius} = \frac{1}{a} \cdot \sqrt{g^2 + f^2 - ac} = \frac{1}{27} \cdot \sqrt{(3)^2 + (-4)^2 - (27)\left(\frac{-2}{3}\right)} = \frac{1}{27} \cdot \sqrt{9 + 16 + 18}$$

$$= \frac{1}{27} \cdot \sqrt{43} = \frac{\sqrt{43}}{27}$$

Example 3:

Find the value of 'a', if $2x^2 + ay^2 - 3x + 2y - 1 = 0$ represents a circle and also find its radius.

Solution :

Since the given equation represents a circle, therefore, coefficient of x^2 = coefficient of y^2

$$\Rightarrow 2 = a$$

Comparing the given circle with $ax^2 + ay^2 + 2gx + 2fy + c = 0$, we have

$$a = 2, 2g = -3, 2f = 2, c = -1 \Rightarrow g = \frac{-3}{2}, f = 1 \text{ and } c = -1$$

$$\therefore \text{Radius} = \frac{1}{a} \cdot \sqrt{g^2 + f^2 - ac}$$

$$= \frac{1}{2} \cdot \sqrt{\left(\frac{-3}{2}\right)^2 + 1^2 - 2(-1)} = \frac{1}{2} \cdot \sqrt{\frac{9}{4} + 1 + 2} = \frac{1}{2} \cdot \sqrt{\frac{21}{4}} = \frac{\sqrt{21}}{4}$$

Example 4:

Find the radius of the circle $\sqrt{1+m^2}(x^2 + y^2) - 2cx - 2mcy = 0$.

[ECET 2012]

Solution :

Comparing the given circle with $ax^2 + ay^2 + 2gx + 2fy + c = 0$, we have

$$a = \sqrt{1+m^2}, g = -c, f = -mc \text{ and } c = 0$$

$$\therefore \text{Radius} = \frac{1}{a} \cdot \sqrt{g^2 + f^2 - ac} = \frac{1}{\sqrt{1+m^2}} \cdot \sqrt{c^2 + m^2c^2} = \frac{c\sqrt{1+m^2}}{\sqrt{1+m^2}} = c$$

Example 5:

If radius of the circle $x^2 + y^2 - 8x + 10y + k = 0$ is 7, then find the value of K.

[ECET 2013]

Solution :

The given circle is $x^2 + y^2 - 8x + 10y + k = 0$

Comparing it with $x^2 + y^2 + 2gx + 2fy + c = 0$, we have

$$2g = -8, 2f = 10, c = k \Rightarrow g = -4, f = 5, c = k$$

Given Radius of the circle, $r = 7$

$$\Rightarrow \sqrt{g^2 + f^2 - c} = 7. \text{ Squaring, } g^2 + f^2 - c = 7^2$$

$$\Rightarrow (-4)^2 + 5^2 - k = 49 \Rightarrow k = 16 + 25 - 49 = -8.$$

Note: Find the centre and radius of the circle $x^2 + y^2 + 8x + 10y - 8 = 0$.

[Oct. 2015 (D.E.E.)]

Example 6:

Find the equation of the circle with center at the point $(2, 3)$ and passing through the point $(-2, -1)$.

Solution :

Let $C(2, 3)$ be the center and $P(-2, -1)$ be a point on the circle.

Then we have

$$CP = \text{radius, } r = \sqrt{(2+2)^2 + (3+1)^2} = \sqrt{16+16} = \sqrt{32}$$

$\therefore$ Equation of the required circle is

$$\begin{aligned} (x-h)^2 + (y-k)^2 &= r^2 \\ \Rightarrow (x-2)^2 + (y-3)^2 &= 32 \Rightarrow x^2 - 4x + 4 + y^2 - 6y + 9 = 32 \\ \Rightarrow x^2 + y^2 - 4x - 6y - 19 &= 0 \end{aligned}$$

Alternate Method :

Given centre, $(h, k) = (2, 3) \Rightarrow h = 2, k = 3$

Let the equation of the circle be

$$(x-h)^2 + (y-k)^2 = r^2$$

Equation of the circle with the given centre $(2, 3)$ is

$$(x-2)^2 + (y-3)^2 = r^2 \quad \dots(1)$$

Since the circle (1) passes through $(-2, -1)$, we have

$$(-2-2)^2 + (-1-3)^2 = r^2 \Rightarrow (-4)^2 + (-4)^2 = r^2$$

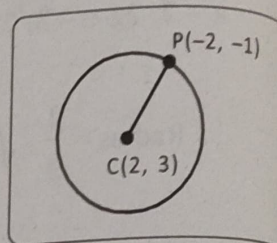
$$\Rightarrow 2(-4)^2 = r^2 \therefore r^2 = 32$$

Putting the value of r^2 in (1), we obtain

$$(x-2)^2 + (y-3)^2 = 32$$

$$\Rightarrow x^2 - 4x + 4 + y^2 - 6y + 9 = 32 \Rightarrow x^2 + y^2 - 4x - 6y - 19 = 0$$

which is the required circle.



Example 7:

Find the equation of the circle concentric with the circle $x^2 + y^2 - 4x - 6y - 9 = 0$ and which passes through the point $(-4, -5)$.

Solution :

We know that Centre of the required circle is the centre of the given circle.

$$\text{Here } 2g = -4, 2f = -6 \Rightarrow g = -2, f = -3$$

$\therefore$ Centre of the given circle is $(-g, -f)$ i.e., $(2, 3)$

Thus centre of the required circle is also $C(2, 3)$

$$\text{i.e., } C(h, k) = C(2, 3)$$

Given point is $P(-4, -5)$

$$\therefore \text{ radius, } r = CP = \sqrt{(2+4)^2 + (3+5)^2} = \sqrt{6^2 + 8^2}$$

$$= \sqrt{36 + 64} = \sqrt{100} = 10$$

Hence equation of the required circle is

$$(x - h)^2 + (y - k)^2 = r^2$$

$$\Rightarrow (x - 2)^2 + (y - 3)^2 = 100 \Rightarrow x^2 - 4x + 4 + y^2 - 6y + 9 = 100$$

$$\Rightarrow x^2 + y^2 - 4x - 6y - 87 = 0$$

Example 8:

The point $(-1, 0)$ lies on the circle $x^2 + y^2 - 4x + 8y + k = 0$. Find the radius of the circle.

[ECET 2015]

Solution :

The point $(-1, 0)$ lies on the circle

$$\Rightarrow (-1)^2 + 0 - 4(-1) + 8(0) + k = 0$$

$$\Rightarrow 1 + 4 + k = 0 \Rightarrow k = -5$$

$$\text{Here } 2g = -4, 2f = 8, c = k = -5$$

$$\Rightarrow g = -2, f = 4, c = -5 (\because k = -5)$$

$$\therefore \text{ Radius} = \sqrt{g^2 + f^2 - c} = \sqrt{(-2)^2 + 4^2 - (-5)}$$

$$= \sqrt{4 + 16 + 5} = \sqrt{25} = 5.$$

Example 9:

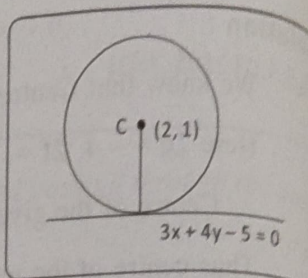
Find the radius of the circle with centre at the point $(2, 1)$ and touches the straight line $3x + 4y - 5 = 0$. [April 2018 (D.C.E)]

Solution :

The radius of the circle, is the perpendicular distance from the centre $(2, 1)$ to the straight line

$$3x + 4y - 5 = 0.$$

$$\therefore \text{Radius, } r = \frac{3(2) + 4(1) - 5}{\sqrt{3^2 + 4^2}} = \frac{5}{\sqrt{25}} = \frac{5}{5} = 1.$$

**Example 10:**

Find the equation of a circle with centre $(4, 1)$ and having $3x + 4y - 1 = 0$ as tangent.

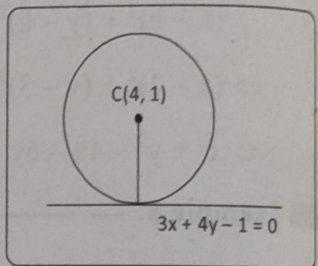
Solution :

Given centre $(h, k) = (4, 1)$. We know that

radius of circle, $r =$ Distance from the centre $(4, 1)$ to the tangent $3x + 4y - 1 = 0$

$$= \frac{|3(4) + 4(1) - 1|}{\sqrt{3^2 + 4^2}} = \frac{12 + 4 - 1}{\sqrt{9 + 16}} = \frac{15}{\sqrt{25}}$$

$$= \frac{15}{5} = 3$$



$$\therefore \text{Equation of the circle is } (x - h)^2 + (y - k)^2 = r^2$$

$$\text{i.e., } (x - 4)^2 + (y - 1)^2 = 3^2 \quad \text{i.e., } x^2 - 8x + 16 + y^2 - 2y + 1 = 9$$

$$\text{or } x^2 + y^2 - 8x - 2y + 8 = 0.$$

Example 11:

Find the equation of the circle having

(i) $(a, 0)$ and $(0, b)$

[April 2015 (D.E.C.E)]

(ii) $(3, 4)$ and $(7, -2)$

(iii) $(1, -5/2)$ and $(-3, -2)$

(iv) $(4, 2)$ and $(1, 5)$

[April 2014 (D.M.E), April 2018 (D.E.E.E)]

as the extremities (end points) of the diameter.

Solution :

(i) Here $(x_1, y_1) = (a, 0)$ and $(x_2, y_2) = (0, b)$

Hence the equation of the required circle is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

$$\text{i.e., } (x - a)(x - 0) + (y - 0)(y - b) = 0$$

$$\text{i.e., } x(x - a) + y(y - b) = 0$$

$$\text{or } x^2 + y^2 - ax - by = 0$$

(ii) We are given $(x_1, y_1) = (3, 4)$ and $(x_2, y_2) = (7, -2)$

Hence equation of the required circle is

$$\text{i.e., } (x - 3)(x - 7) + (y - 4)(y + 2) = 0$$

$$\text{i.e., } (x^2 - 10x + 21) + (y^2 - 2y - 8) = 0$$

$$\text{or } x^2 + y^2 - 10x - 2y + 13 = 0.$$

This is the required circle.

Alternate Method :

Let A (3, 4), B (7, -2) be the end points of the diameter.

∴ Centre, C = Mid-point of AB

$$= \left(\frac{3+7}{2}, \frac{4-2}{2} \right) = (5, 1)$$

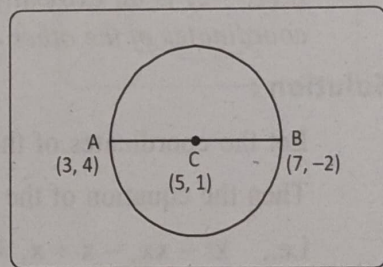
i.e., $(h, k) = (5, 1)$. Now radius, $r = AC$

$$\therefore r^2 = AC^2 = (3 - 5)^2 + (4 - 1)^2 = 4 + 9 = 13$$

Hence equation of the required circle is $(x - h)^2 + (y - k)^2 = r^2$

$$\text{i.e., } (x - 5)^2 + (y - 1)^2 = 13$$

$$\text{or } x^2 + y^2 - 10x - 2y + 13 = 0.$$



Note: Students are advised that find the equation of the circle using the formula only.

(iii) This is left as an exercise to the reader. Proceeding as above, we get the equation of the circle as $2x^2 + 2y^2 + 4x + 9y + 4 = 0$.

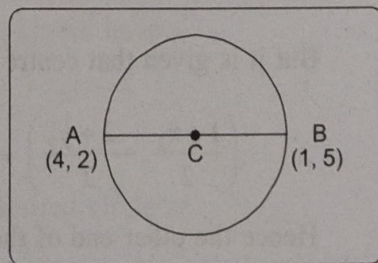
(iv) **Method 1 :** This is left as an exercise to the reader.

Method 2 : Without using the formula.

Let A (4, 2) and B(1, 5) be the end points of the diameter.

Then centre of the circle = Mid-point of AB

$$= \left(\frac{4+1}{2}, \frac{2+5}{2} \right) = \left(\frac{5}{2}, \frac{7}{2} \right)$$



$$\text{i.e., } C(h, k) = \left(\frac{5}{2}, \frac{7}{2}\right)$$

Now radius of the circle, $r = AC$ (or CB)

$$\begin{aligned}\therefore r^2 &= AC^2 = \left(\frac{5}{2} - 4\right)^2 + \left(\frac{7}{2} - 2\right)^2 \\ &= \left(\frac{5-8}{2}\right)^2 + \left(\frac{7-4}{2}\right)^2 = \left(\frac{-3}{2}\right)^2 + \left(\frac{3}{2}\right)^2 = \frac{9}{4} + \frac{9}{4} = 2\left(\frac{9}{4}\right) = \frac{9}{2}\end{aligned}$$

Hence equation of the required circle is

$$(x - h)^2 + (y - k)^2 = r^2$$

$$\text{i.e., } \left(x - \frac{5}{2}\right)^2 + \left(y - \frac{7}{2}\right)^2 = \frac{9}{2}$$

Example 12:

If $(1, -2)$ is an extremity of a diameter of the circle $x^2 + y^2 - 4x + 6y + 13 = 0$, find the coordinates of the other extremity of the diameter.

Solution :

Let the coordinates of the other extremity be (x_1, y_1) .

Then the equation of the circle is $(x - 1)(x - x_1) + (y + 2)(y - y_1) = 0$

$$\text{i.e., } x^2 - xx_1 - x + x_1 + y^2 - yy_1 + 2y - 2y_1 = 0$$

$$\text{i.e., } x^2 + y^2 - x(1 + x_1) + y(2 - y_1) + x_1 - 2y_1 = 0$$

Comparing it with the circle $x^2 + y^2 - 4x + 6y + 13 = 0$, we get

$$x_1 + 1 = 4, \quad 2 - y_1 = 6 \Rightarrow x_1 = 3, \quad y_1 = -4$$

$\therefore$ The other extremity of the diameter is $(3, -4)$.

An Easiest Method :

$$\text{Here } 2g = -4, \quad 2f = 6 \Rightarrow g = -2, \quad f = 3.$$

$$\therefore \text{ The centre of the given circle, } C = (-g, -f) = (2, -3)$$

Let $A(1, -2)$ and $B(x_2, y_2)$ be the ends of a diameter of the given circle. Then

$$\text{Centre } C \text{ of the circle} = \text{Mid-point of } AB = \left(\frac{1+x_2}{2}, \frac{-2+y_2}{2}\right)$$

But it is given that centre of the circle is $(2, -3)$

$$\therefore \left(\frac{1+x_2}{2}, \frac{-2+y_2}{2}\right) = (2, -3) \Rightarrow \frac{1+x_2}{2} = 2, \quad \frac{-2+y_2}{2} = -3 \Rightarrow x_2 = 3, \quad y_2 = -4$$

Hence the other end of the diameter is $(3, -4)$.

Another Method: Using the Formula

Given one extremity, $(x_1, y_1) = (1, -2)$ and $2g = -4, 2f = 6$.

∴ The other extremity, $(x_2, y_2) = (-2g - x_1, -2f - y_1) = (4 - 1, -6 + 2) = (3, -4)$.

Example 13:

If one end of the diameter of the circle $x^2 + y^2 - 5x - 8y + 13 = 0$ is $(2, 7)$, then find the other end of the diameter. [ECET 2012]

Solution :

Let $A(2, 7)$ and $B(x, y)$ be the ends of a diameter of the circle $x^2 + y^2 - 5x - 8y + 13 = 0$.

$$\text{Here } 2g = -5, 2f = -8 \Rightarrow g = \frac{-5}{2}, f = -4$$

$$\therefore \text{Centre of the circle, } C = (-g, -f) = \left(\frac{5}{2}, 4\right)$$

We have centre C of the circle = mid-point of AB

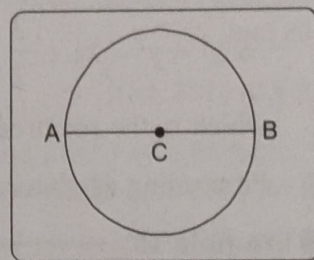
$$\Rightarrow \left(\frac{5}{2}, 4\right) = \left(\frac{2+x}{2}, \frac{7+y}{2}\right)$$

$$\Rightarrow \frac{5}{2} = \frac{2+x}{2} \text{ and } 4 = \frac{7+y}{2}$$

$$\Rightarrow 5 = 2 + x \text{ and } 7 + y = 8$$

$$\Rightarrow x = 3, y = 1$$

∴ The other end is $(3, 1)$

**Example 14:**

Find the equation of the circle passing through the points

(i) $(0, 0)$, $(1, 0)$ and $(0, 1)$

(ii) $(0, 0)$, $(1, -2)$ and $(2, 0)$

(iii) $(0, 0)$, $(a, 0)$ and $(0, a)$

Solution :

(i) Let the equation of the required circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \dots (1)$$

Since it is passing through the points $(0, 0)$, $(1, 0)$ and $(0, 1)$, we have

$$c = 0 \quad \dots (2)$$

$$1 + 2g + c = 0 \Rightarrow 2g = -1, \text{ using (2)}$$

$$\text{and } 1 + 2f + c = 0 \Rightarrow 2f = -1, \text{ using (2)}$$

Substituting the values of $2g$, $2f$ and c in (1), we get the desired circle as

$$x^2 + y^2 - x - y = 0.$$

(ii) Let the equation of the required circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \dots (1)$$

Since it is passing through the points (0, 0), (1, -2) and (2, 0), we have

$$c = 0 \quad \dots (2)$$

$$1 + 4 + 2g - 4f + c = 0 \Rightarrow 2g - 4f = -5 \quad \dots (3) \quad [\because c = 0]$$

$$4 + 0 + 4g + 0 + c = 0 \Rightarrow 4g = -4 \quad [\because c = 0] \therefore g = -1$$

$$\text{From (3), } 4f = 2g + 5 = 2(-1) + 5 = -2 + 5 = 3 \Rightarrow f = \frac{3}{4}$$

Substituting the values of g, f and c in (1), we get

$$x^2 + y^2 + 2(-1)x + 2\left(\frac{3}{4}\right)y = 0$$

$$\Rightarrow x^2 + y^2 - 2x + \frac{3}{2}y = 0 \Rightarrow 2x^2 + 2y^2 - 4x + 3y = 0$$

which is the required circle.

(iii) Proceeding as above, we get the equation of the required circle as $x^2 + y^2 - ax - ay = 0$.

Example 15:

Find the equation of the circle passing through the points (1, 2), (3, -4) and (5, -6).

Solution :

Let the equation of the required circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ (1)

Since it passes through (1, 2), (3, -4) and (5, -6), we have

$$5 + 2g + 4f + c = 0 \quad \dots (2)$$

$$25 + 6g - 8f + c = 0 \quad \dots (3)$$

$$61 + 10g - 12f + c = 0 \quad \dots (4)$$

(3) - (2) gives

$$4g - 12f + 20 = 0 \quad \dots (5)$$

(4) - (3) gives

$$4g - 4f + 36 = 0 \quad \dots (6)$$

(5) - (6) gives

$$-8f - 16 = 0 \Rightarrow f = -2$$

$$\text{From (5), } 4g = 12f - 20 = -24 - 20 = -44 \therefore g = -11$$

$$\text{From (2), } c = -5 - 2g - 4f = -5 + 22 + 8 = 25$$

Substituting the values of g, f and c in (1), we get the required equation to the circle as $x^2 + y^2 - 22x - 4y + 25 = 0$.

Review Questions

1. Define circle.
 2. Define point circle and concentric circles. [April 2018 (D.E.E.E)]
 3. Write the equation of circle whose centre is (h, k) and radius is r .
[April 2016 (D.E.E.E), 2018 (D.E.C.E, Common), 2019, Sept. 2021, Jan. 2024 (MID-I)]
 4. Write the general equation of a circle and mention its centre and radius.
[Oct. 2016 (D.E.C.E)]
 5. Write the equation of the point circle whose centre is (h, k) .
 6. Write the formula to find the centre and radius of the circle
 - (i) $x^2 + y^2 + 2gx + 2fy + c = 0$ [April 2017 (A/AA/AEI...), April 2018 (D.C.E), Jan. 2023]
 - (ii) $ax^2 + ay^2 + 2gx + 2fy + c = 0$ [Oct. 2016 (D.E.E.E)]
 7. Write the equation of circle having (x_1, y_1) and (x_2, y_2) as end points of a diameter.
[May 2022, Jan. 2025 (MID-I)]
- (or) Write the equation of a circle whose end points of a diameter are (x_1, y_1) and (x_2, y_2) .

Objective Questions

I. Fill in the Blanks with Correct Answers :

1. Equation of a point circle with centre (h, k) is _____.
2. Equation of a point circle with centre $(0, 0)$ is _____.
3. The centre of the circle $x^2 + y^2 - 2x - 4y - 11 = 0$ is _____.
4. The radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ is _____.

II. State True/False :

1. The centre of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is (g, f) .
2. The centre of the circle $ax^2 + ay^2 + 2gx + 2fy + c = 0$ is $\left(\frac{-g}{a}, \frac{-f}{a}\right)$.
3. The radius of the circle $ax^2 + ay^2 + 2gx + 2fy + c = 0$ is $\frac{1}{a} \cdot \sqrt{g^2 + f^2 - ac}$.

4. Equation of the circle with centre origin and radius r is $x^2 + y^2 = r^2$.
5. The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a real circle if $(g^2 + f^2 - c) > 0$.
6. The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a point circle if $(g^2 + f^2 - c) < 0$.

Answers

- I. 1. $(x - h)^2 + (y - k)^2 = 0$ 2. $x^2 + y^2 = 0$ 3. $(1, 2)$ 4. 5
- II. 1. False 2. True 3. True 4. True 5. True
6. False

Exercise 2.1

1. (i) Find the equation to the circle

(a) whose centre is $(-1, 2)$ and radius is 3

(b) whose centre is $(2, 5)$ and radius is 2

[Oct. 2016 (D.M.E)]

(c) whose centre is $(a, -b)$ and radius is $a + b$

(d) whose centre is $(1, 2)$ and radius is 5. [April 2016 (D.M.E), 2018 (D.M.E, D.C.M.E)]

(e) whose centre is $(2, 2)$ and radius is 2.

- (ii) Find the equation of the point circle with centre

(a) $(9, -2)$ [Oct. 2015 (D.M.E, D.C.E), April 2019 (C-16S)]

(b) $(11, 3)$

(c) $(3, -7)$ [April 2014 (D.C.E)]

(d) $(2, -3)$ [April 2017 (D.C.E)]

(e) $(3, 5)$ [April 2017 (D.E.E.E), Jan. 2025 (MID-I)]

(f) $(13, -9)$ [April 2018 (D.C.E)]

2. (i) Find the equation of the circle whose centre is $(2, 3)$ and which passes through the origin.
- (ii) Find the equation of the circle whose centre is $(-1, 2)$ and which passes through $(5, 6)$. [Sept. 2021]
- (iii) Find the equation of the circle passing through $(2, -1)$ and having the centre at $(2, 3)$
- (iv) Find the equation of the circle passing through $(-7, 1)$ and having centre at $(-4, -3)$

- (v) Find the equation of the circle passing through (4, 5) having the centre at (2, 2).
 (vi) Find the area of the circle with centre (2, 2) and passing through (4, 6).
 (vii) Find the area of the circle $(x + 1)(x + 2) + (y - 1)(y + 3) = 0$.

3. (i) Find the equation of the circle which has its centre at the point (6, 1) and touches the straight line $5x + 12y - 3 = 0$.
 (or) Find the radius of the circle which has its centre at the point (6, 1) and touches the line $5x + 12y - 3 = 0$. Also find equation of the circle.
 (ii) Find the equation of the circle having the centre at (3, 4) and touching the straight line $5x + 12y - 11 = 0$. [Jan. 2025 (MID-I)]
 (iii) Find the equation of the circle which has its centre at the point (2, 1) and touches the straight line $3x + 4y - 5 = 0$. [Sept. 2021]

[Hint : Refer Solved Example 9]

4. Find the centre and radius of the circle

- (i) (a) $x^2 + y^2 + 4x - 6y = 0$ [April 2016 (D.E.C.E, D.E.E.E), 2017 (D.E.C.E), April 2018 (D.E.E.E), 2019 (C-16S)]
 (b) $x^2 + y^2 + 4x - 8y = 0$ [Oct. 2016 (D.E.E.E), Oct. 2020 (Common)]
 (ii) (a) $x^2 + y^2 - 6x + 4y - 12 = 0$ [Oct. 2016 (A/AA/AEI...), April 2018 (D.CM.E), Nov. 2021 (Sup), Jan. 2023 (MID-I)]
 (b) $x^2 + y^2 - 4x - 6y - 12 = 0$. [April 2018 (D.M.E)]
 (iii) $3x^2 + 3y^2 - 12x + 6y + 11 = 0$ [Oct. 2007, April 2008 (DECE), April 2009 (DCE), Oct. 2010, Oct. 2015 (D.M.E), April 2017 (D.E.E.E)]
 (iv) $5x^2 + 5y^2 - 2x - 3y = 0$
 (v) $5x^2 + 5y^2 - 20x - 30y - 1 = 0$
 (vi) $2x^2 + 2y^2 + 3x + 7y + 2 = 0$.

5. Find the equation of the circle with

- (i) (2, 3) and (6, 9) [April 2015 (D.E.E.E), Oct. 2015 (D.E.C.E), 2017 (D.C.E, D.E.C.E), April 2018 (D.M.E, D.E.C.E, Common), April 2019 (C-16S)]
 (ii) (-5, 1) and (3, -7) [April 2015 (Common)]
 (iii) (3, -4) and (-2, 5)
 (iv) (1, 2) and (4, 5)
 (v) (2, 4) and (1, 3). [April 2017 (A/AA/AEI...)]
 (vi) (2, 0) and (0, 4). [April 2019]

(vii) (2, 1) and (-4, 3).

(viii) (1, -1) and (2, 3).

(ix) (0, 1) and (3, 0).

(x) (1, 2) and (2, 1).

as the end points of a diameter.

6. (i) If (2, 1) is an extremity of a diameter of the circle $3x^2 + 3y^2 - 9x + 15y - 7 = 0$, find coordinates of the other extremity of the diameter.

- (ii) If (3, 4) is one end of a diameter of the circle $x^2 + y^2 - 4x - 6y + 11 = 0$, find the other end of the diameter. [Oct. 2020 (Common)]

7. Find the equation of the circle passing through the points

(i) (a) (0, 0), (a, 0) and (0, b) [April 2017 (D.E.E.E)]

(b) (0, 0), (6, 0), (0, 8) [April 2018 (D.E.C.E, D.CM.E, Common), 2019 (C-16S), 2020 (C-16S)]

(c) (0, 0), (1, 0) and (0, 2)

(ii) (0, 0), (6, 0) and (8, 4) [April 2017, 2018 (D.M.E)]

(iii) (1, 0), (0, -6) and (3, 4) [April 2018 (D.C.E)]

(iv) (0, 1), (2, 3) and (-2, 5)

(v) (1, 1), (2, -1) and (3, 2)

(vi) (-2, 8), (7, -1) and (1, 2).

(vii) (5, 2), (2, 1) and (0, 0)

(viii) (0, 0), (0, 6) and (3, 4) [April 2018 (D.E.E.E)]

Answers

1. (i) (a) $x^2 + y^2 + 2x - 4y - 4 = 0$ (b) $x^2 + y^2 - 4x - 10y + 25 = 0$
 (c) $x^2 + y^2 - 2ax + 2by - 2ab = 0$ (d) $x^2 + y^2 - 2x - 4y - 20 = 0$
 (e) $x^2 + y^2 - 4x - 4y + 4 = 0$
 (ii) (a) $x^2 + y^2 - 1x + 4y + 85 = 0$ (b) $x^2 + y^2 - 22x - 6y + 130 = 0$
 (c) $x^2 + y^2 - 6x + 14y + 58 = 0$ (d) $x^2 + y^2 - 4x + 6y + 13 = 0$
 (e) $x^2 + y^2 - 6x - 10y + 34 = 0$ (f) $x^2 + y^2 - 26x + 18y + 250 = 0$
 (iv) $\frac{17\pi}{2}$
2. (i) $x^2 + y^2 - 4x - 6y = 0$ (ii) $x^2 + y^2 + 2x - 4y - 47 = 0$
 (iii) $x^2 + y^2 - 4x - 6y - 3 = 0$ (iv) $x^2 + y^2 + 8x + 6y = 0$
 (v) $x^2 + y^2 - 4x - 4y - 5 = 0$ (vi) 20π sq.units
3. (i) $x^2 + y^2 - 12x - 2y + 28 = 0$ (ii) $x^2 + y^2 - 6x - 8y + 9 = 0$
 (iii) $x^2 + y^2 - 4x - 2y + 4 = 0$
4. (i) (a) $(-2, 3); \sqrt{13}$ (b) $(-2, 4); 2\sqrt{5}$ (ii) (a) $(3, -2); 5$ (b) $(2, 3); 5$
 (iii) $(2, -1); 2/\sqrt{3}$ (iv) $\left(\frac{1}{5}, \frac{3}{10}\right); \frac{\sqrt{13}}{10}$
 (v) $(2, 3); \sqrt{66/5}$ (vi) $\left(-\frac{3}{4}, -\frac{7}{4}\right); \frac{\sqrt{42}}{4}$
5. (i) $x^2 + y^2 - 8x - 12y + 39 = 0$ (ii) $x^2 + y^2 + 2x + 6y - 22 = 0$
 (iii) $x^2 + y^2 - x - y - 26 = 0$ (iv) $x^2 + y^2 - 5x - 7y + 14 = 0$
 (v) $x^2 + y^2 - 3x - 7y + 14 = 0$ (vi) $x^2 + y^2 - 2x - 4y = 0$
 (vii) $x^2 + y^2 + 2x - 4y - 5 = 0$ (viii) $x^2 + y^2 - 3x - 2y - 1 = 0$
 (ix) $x^2 + y^2 - 3x - y = 0$ (x) $x^2 + y^2 - 3x + 3y + 4 = 0$
6. (i) $(1, -6)$ (ii) $(1, 2)$
7. (i) (a) $x^2 + y^2 - ax - by = 0$
 (b) $x^2 + y^2 - 6x - 8y = 0$
 (c) $x^2 + y^2 - x - 2y = 0$
 (ii) $x^2 + y^2 - 6x - 13y = 0$ (iii) $4x^2 + 4y^2 - 142x + 47y + 138 = 0$
 (iv) $3x^2 + 3y^2 + 2x - 20y + 17 = 0$ (v) $x^2 + y^2 - 5x - y + 4 = 0$
 (vi) $x^2 + y^2 - 17x - 19y + 50 = 0$ (vii) $x^2 + y^2 - 19x + 33y = 0$

2.7 TANGENTS AND NORMALS

If a circle and a line L are drawn in a plane, then only the following three situations are possible.

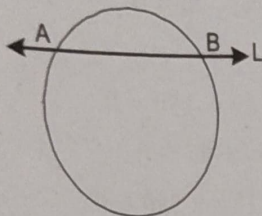


Fig. 2.6

The line intersects the circle at two points A and B

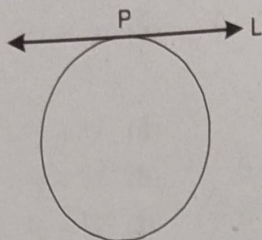


Fig. 2.7

The line intersects the circle at only one point P

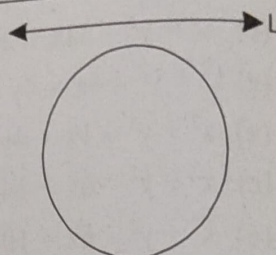


Fig. 2.8

The line does not intersect the circle.

Definitions :

1. A line which intersects the circle in two points is called a *secant* of the circle. That is, if A and B are any two points on a circle, then the line $\overleftrightarrow{AB}$ passing through A and B is called a *secant* line and the line segment $\overline{AB}$ joining A and B is called a *chord*.

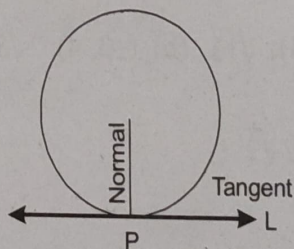
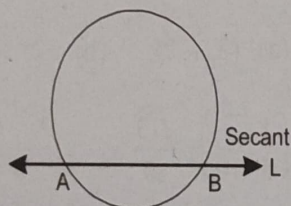


Fig. 2.9

2. A line which intersects the circle in exactly one point is called a *tangent*. The point of intersection, P, is called the *point of contact*, or the *point of tangency*. A circle and a tangent are said to touch each other at the point of contact.
3. The line through the point of contact P and perpendicular to the tangent is called the *normal*.

An Alternative Definition :

Tangent and Normal to a curve at a given point :

Let P be a given point and Q any other point on it. As Q approaches P then the chord PQ tends to a definite straight line PT which is called the tangent to the curve at P. That is, the limiting position of the chord PQ when $Q \rightarrow P$ along the curve is called the tangent to the curve at P. The point P is called the point of contact.

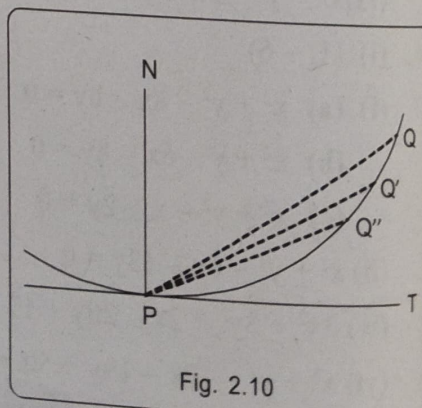


Fig. 2.10

The straight line PN, through P perpendicular to the tangent PT, is called the normal to the curve at P.

Note: The tangent at any point of a circle and the radius through this point are perpendicular to each other.

2.8.1 Theorem

Equation of the tangent to the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ at the point $P(x_1, y_1)$ on it is $S_1 \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$.

Proof :

Let C be the centre of the circle and $P(x_1, y_1)$ be a point on it. Join CP. From P draw $PB \perp$ to CP. Then PB is the tangent at $P(x_1, y_1)$.

$$\text{Slope of CP} = \frac{y_1 - (-f)}{x_1 - (-g)} = \frac{y_1 + f}{x_1 + g}$$

Since PB is $\perp$ to CP, therefore,

$$\text{Slope of PB is } \frac{-(x_1 + g)}{y_1 + f}$$

Equation to the tangent PB is

$$y - y_1 = m(x - x_1)$$

$$\text{i.e., } y - y_1 = \frac{-(x_1 + g)}{y_1 + f}(x - x_1)$$

$$\text{i.e., } (x_1 + g)(x - x_1) + (y_1 + f)(y - y_1) = 0$$

$$\text{i.e., } xx_1 + yy_1 + gx + fy = x_1^2 + y_1^2 + gx_1 + fy_1$$

Adding $gx_1 + fy_1 + c$ on both sides, we obtain

$$xx_1 + yy_1 + gx + fy + gx_1 + fy_1 + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c \quad \dots (1)$$

Since (x_1, y_1) lies on the circle, we have

$$x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0 \quad \dots (2)$$

Using (2) in (1), we get

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

This is the equation of the tangent to the given circle at the point (x_1, y_1) on it.

Note : Equation of the tangent to the circle $x^2 + y^2 = a^2$ at $P(x_1, y_1)$ is $xx_1 + yy_1 = a^2$. This is left as an exercise to the student.

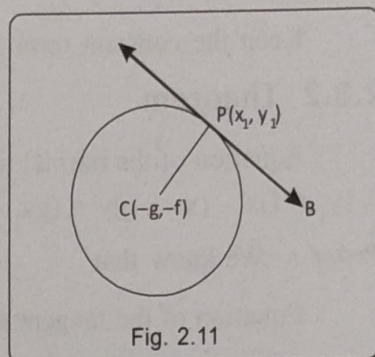


Fig. 2.11

Rule to write the equation of the tangent at $P(x_1, y_1)$.

For x^2 , write xx_1

For y^2 , write yy_1

For x , write $\frac{1}{2}(x + x_1)$

For y , write $\frac{1}{2}(y + y_1)$

Keep the constant term unchanged i.e., for c , write c .

2.8.2 Theorem

Equation of the normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at the point $P(x_1, y_1)$ on is $(y_1 + f)x - (x_1 + g)y + (gy_1 - fx_1) = 0$ [(or) $(y_1 + f)(x - x_1) - (x_1 + g)(y - y_1) = 0$].

Proof : We know that

Equation of the tangent to the given circle at (x_1, y_1) is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$\text{i.e., } x(x_1 + g) + y(y_1 + f) + (gx_1 + fy_1 + c) = 0$$

$$\text{Slope of the tangent} = -\frac{\text{coefficient of } x}{\text{coefficient of } y} = -\frac{x_1 + g}{y_1 + f}$$

Since normal is always perpendicular to tangent,

$$\therefore \text{Slope of the normal} = \frac{y_1 + f}{x_1 + g}$$

Hence equation of the normal at (x_1, y_1) is

$$y - y_1 = m(x - x_1)$$

$$\text{i.e., } y - y_1 = \frac{y_1 + f}{x_1 + g}(x - x_1)$$

$$\text{i.e., } (y - y_1)(x_1 + g) = (y_1 + f)(x - x_1)$$

$$\text{or } (x_1 + g)y - x_1 y_1 - gy_1 = (y_1 + f)x - x_1 y_1 - fx_1$$

$$\text{or } (y_1 + f)x - (x_1 + g)y + (gy_1 - fx_1) = 0$$

Note : Equation of the normal to the circle $x^2 + y^2 = a^2$ at the point (x_1, y_1) is

$$y - y_1 = \frac{y_1}{x_1}(x - x_1) \text{ (or) } y_1 x - x_1 y = 0.$$

Solved Examples

Example 1:

Find the equation of the tangent and normal to the circle

$$x^2 + y^2 - 10x + 4y - 140 = 0 \text{ at } (-7, -7) \text{ on it.}$$

[April 2017 (D.E.C.E), Jan. 2024 (MID-I)]

Solution :

Comparing the given circle with standard form, we have

$$g = -5, f = 2, c = -140$$

$$\text{Also } (x_1, y_1) = (-7, -7)$$

$\therefore$ Equation of the tangent is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$\text{i.e., } x(-7) + y(-7) - 5(x - 7) + 2(y - 7) - 140 = 0$$

$$\text{or } 12x + 5y + 119 = 0.$$

$$\text{Now slope of the tangent} = m = -\frac{\text{coefficient of } x}{\text{coefficient of } y} = \frac{-12}{5}$$

$\therefore$ Equation of the normal at $P(-7, -7)$ is

$$y - y_1 = -\frac{1}{m}(x - x_1)$$

$$\text{i.e., } y + 7 = \frac{5}{12}(x + 7)$$

$$\text{i.e., } 12y + 84 = 5x + 35$$

$$\text{or } 5x - 12y - 49 = 0.$$

Example 2:

Find the equations of the tangents to the circle $x^2 + y^2 - 4x - 6y + 4 = 0$ which are parallel to $3x - 4y - 17 = 0$.

Solution :

The equation of the tangent parallel to the given line is $3x - 4y + k = 0$. Centre of the given circle is $(2, 3)$ and its radius $= \sqrt{g^2 + f^2 - c} = \sqrt{4 + 9 - 4} = 3$.

Since the line (1) is a tangent to the circle, we have

Length of perpendicular from the centre to the line = radius.

$$\Rightarrow \left| \frac{6 - 12 + k}{\sqrt{9 + 16}} \right| = 3 \Rightarrow \left| \frac{k - 6}{5} \right| = 3$$

$$\text{i.e., } \frac{k - 6}{5} = \pm 3 \quad \text{i.e., } k - 6 = \pm 15 \Rightarrow k = 6 \pm 15.$$

This gives $k = 6 + 15 = 21$, $k = 6 - 15 = -9$.

Hence the equations of tangents parallel to the given line are $3x - 4y + 21 = 0$ and $3x - 4y - 9 = 0$.

Review Questions

1. Define tangent line to the circle (or) curve.
2. Write the equation of tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at (x_1, y_1) .
3. Write the equation of tangent to the circle $x^2 + y^2 = r^2$ at (x_1, y_1) .
[Oct. 2016 (A/AA/CH...), April 2018 (D.C.M.E)]

Exercise 2.2

1. Find the equations of the tangent and normal to the circle
 - (i) $x^2 + y^2 - 6x - 3y - 2 = 0$ at $(2, -2)$.
 - (ii) $2x^2 + 2y^2 + 3x - 4y + 1 = 0$ at $(-1, 2)$.
 - (iii) $3x^2 + 3y^2 - 19x - 29y + 76 = 0$ at $(1, 3)$.
 - (iv) $x^2 + y^2 - 6x + 4y - 4 = 0$ at $(4, 2)$.
[April 2018 (D.M.E), Jan. 2023 (MID-I)]

Answers

1.
 - (i) $2x + 7y + 10 = 0$; $7x - 2y - 18 = 0$.
 - (ii) $x - 4y + 7 = 0$; $4x + y + 2 = 0$.
 - (iii) $11x - 13y + 28 = 0$.
 - (iv) $x + 4y - 12 = 0$; $4x - y - 14 = 0$.